

BST as a Causal Set: Causality as the Derived Commitment Order on D_{IV}^5

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Abstract

We give the causal-structure formulation of Bubble Spacetime Theory (BST). The result is stated plainly: BST's causal structure is a causal set — a locally finite partial order — in the precise sense of Bombelli-Lee-Meyer-Sorkin (1987). The partial order is the commitment order of D_{IV}^5 's commit operator: event a precedes event b iff a 's contact commits before b 's in the substrate's sequential growth. Comparable pairs are timelike-related; incomparable pairs are spacelike; the growth step is the arrow of time. The decisive feature — the one standard causal-set theory *lacks* — is that BST does not posit or randomly grow the order; it derives it from the geometry. The *tangent-space* signature is fixed to (3,1) (Structure-Derived, T2545), and *Order + Number = Geometry* (Malament + Sorkin) is realized literally, with BST supplying both the order (commit sequence) and the number (commitment count = Casimir energy). BST realizes causality by a different, order-theoretic route: where a Causal Fermion System *reconstructs* causality from an internal spinor-Krein spectral object (the A_{xy} classification) plus a causal-action minimization, BST's causality is the order directly — and BST does not satisfy Finster's specific A_{xy} classification (a structural difference, not a ranking). The one genuinely open theorem — the continuum limit — is stated honestly and split into a part now shown and a part still open: the commit-forced order escapes the Kleitman-Rothschild pancake (the measure-zero trap of generic posets — at the physical size $N=137$ its Myrheim-Meyer ordering fraction is 0.098 vs the KR pancake's 0.626, a clean $\sim 6x$ separation), which is real progress; but the continuum-limit dimension and metric remain open and unmeasured — escaping KR is *not* a dimension measurement. BST enters with the tangent signature and volume measure pinned and the KR trap escaped; the continuum dimension is the honest open frontier.

1. The commitment order

BST's substrate operates by *commitment*: contacts on the Shilov boundary of D_{IV}^5 commit in a definite sequence, driven by the commit operator (the $K = SO(5) \times SO(2)$ Casimir generating the $SO(2)$ time-circle evolution). Define the relation on commitment events:

$a \prec b$ iff a lies in the conformal (Lorentzian) causal past of b on the Shilov boundary — grounded non-circularly on the boundary's conformal structure, not on the commitment dynamics it defines.

Concretely the Shilov boundary $\hat{S} = S^4 \times S^1 / Z_2$ carries a conformal Lorentzian structure; unwrapping the $SO(2)$ time-circle $S^1 \rightarrow \mathbb{R}$ gives $\mathbb{R} \times S^4$ (4+1-dimensional: S^4 = space, $\mathbb{R}$ = time), and $\prec$ is its conformal causal order — a is in the causal past of b .

This relation is: - reflexive ($a \prec a$), - antisymmetric ($a \prec b$ and $b \prec a \Rightarrow a = b$ — no closed commitment loops; the $SO(2)$ phase advances monotonically once unfrozen), - transitive ($a \prec b \prec c \Rightarrow a \prec c$),

so (commitment events, $\prec$) is a partial order. It is locally finite: between any two commitments only finitely many commitments intervene (the substrate clock is discrete — the Koons tick). A locally finite partial order is,

by definition, a causal set.

Comparable iff timelike; incomparable iff spacelike. Two commitments that stand in the order (one in the other's past) are timelike-related; two that are order-incomparable are spacelike — they commit without either preceding the other. The arrow of time is the growth direction: the “march” by which the causal set accretes new commitments (Rideout-Sorkin classical sequential growth is the generic form; BST's growth is the *specific*, geometry-fixed one).

2. BST satisfies the causal-set axioms

BLMS axiom	BST realization
Partial order (reflexive, antisymmetric, transitive)	commitment order $-<$ (Section 1)
Local finiteness (finite interval cardinality) — the <i>dynamics</i> / <i>why this order</i> —	discrete commitment clock (Koons tick) derived from D_{IV}^5 's commit operator

The first two rows make BST a causal set. The third row is where BST departs from — and improves on — standard causal-set theory.

3. The order is derived, not posited (the decisive advantage)

Standard causal-set theory faces the problem of the dynamics: the causal order is either put in by hand or generated by *classical sequential growth* with free parameters (the Rideout-Sorkin coupling constants t_n) tuned so the growth is covariant and Markovian. The theory does not say *which* causal set the universe is, nor *why*.

BST removes this freedom. The commitment order is forced by the commit operator on D_{IV}^5 : the $SO(2)$ time-circle evolution of the K-Casimir dictates the sequence in which boundary contacts commit. There are no growth parameters to tune — the five integers fix the geometry, the geometry fixes the commit operator, and the commit operator fixes the order. BST is the causal set whose growth law is derived rather than chosen. This is precisely the piece causal-set theory has spent decades seeking.

4. Order + Number = Geometry (realized literally)

The Malament-Hawking-King-McCarthy theorem: for a spacetime of dimension > 2 , the causal order determines the metric up to a conformal factor. Sorkin's slogan completes it — the missing conformal factor is volume, and on a causal set volume is counting (number of elements). Hence *Order + Number = Geometry*.

BST supplies both ingredients from one structure: - Order — the commitment partial order (Section 1). - Number — the commitment count, which BST already identifies with the Casimir / commit energy (the volume measure on the substrate). Counting commitments *is* measuring spacetime volume.

So BST does not merely *have* a causal order; it comes with the canonical volume measure that turns order into full metric geometry — the two data Malament + Sorkin require, both derived, neither fit.

5. Signature and frames

Tangent-space signature (3,1) is Structure-Derived (T2545). The three spatial directions are the off-diagonal Peirce component $V_{12} = J_{\{1/2\}}$ of the *simple* D_{IV}^5 spin factor, of dimension $n_C - 2 = 3 = N_c$; by Jordan structure theory a simple algebra's off-diagonal Peirce space is irreducible (the $SO(3)$ vector / $SU(3)$ fundamental), so no

proper invariant subspace splits the three — they come as one triplet, plus the single commit/time direction: $3 + 1$. This is the signature of the tangent algebra, and it is what is pinned. It is *not* the same as the continuum-limit dimension of the causal set, which is a separate, open and unmeasured quantity (Section 8). Conflating “the tangent signature is (3,1)” with “the continuum limit is 4-dimensional” would assume half of the open theorem — we keep them distinct.

Frames and Lorentz invariance (resolving the “totalizer” tension, K1227). The commit *energy* (Casimir) defines a time function — a linear extension of the partial order, i.e. a foliation / frame. It is not the causal order itself. The causal order is the frame-invariant partial dominance order; a frame’s ties (commit-energy-equal events) are its simultaneity slice; a Lorentz boost is a *different admissible time function* of the *same* order. Totalizing the order = *choosing a frame*, not destroying space. Thus the causal structure is Lorentz-invariant at the level of the order, and boosts permute the admissible linear extensions — exactly the causal-set account of relativity.

6. Why BST is a causal set and not a Causal Fermion System (the mechanism)

We asked whether BST’s fermionic sector is a Finster Causal Fermion System (CFS) — whether the BST two-point kernel reproduces Finster’s causal classification via the spectral object $A_{xy} = P(x,y)P(y,x)$. The answer is a clean structural no (the two-point Krein symmetry carries an irreducible 0.465 gap; no fundamental symmetry J can be simultaneously the *geometric* signature and the *internal* spinor signature). This negative is the positive result of this paper, and the reason is sharp:

BST’s indefiniteness is *geometric*, not *internal*. The Lorentzian/causal signature lives in the order — it *moves points* (which commitment precedes which). It does not live in an internal spinor-Krein spectral structure. Finster’s construction *manufactures* causality from an internal object (the fermionic projector’s A_{xy} eigenvalues, equal iff spacelike, distinct iff timelike). BST has no need to manufacture it: causality is already there, in the order commitments happen.

Causal sets are a *substructure* of causal fermion systems (CFS is the fermionic upgrade). BST sits at the causal-set level directly — it realizes the causal order without the fermionic overhead. The G2 negative is exactly the statement that BST is the smaller, more fundamental object.

7. Simpler than Finster (the comparison, stated fairly)

	Finster (CFS)	BST
Where causality lives	reconstructed from internal spinor-Krein A_{xy} spectrum	the commitment order itself
Required apparatus	internal (2,2) spinor-Krein structure + A_{xy} construction + causal-action minimization	the partial order, full stop
Dynamics	minimize the causal action functional	derived commit operator (no functional to minimize)
Tangent signature	emergent, model-dependent	(3,1) Structure-Derived (T2545) — continuum-limit dimension distinct and unmeasured
Volume / number	causal-action weight	commitment count = Casimir energy

This is not a claim that Finster is wrong. It is that BST realizes causality by a different, simpler, self-standing route — the order — where Finster reconstructs it spectrally. Both are legitimate; BST’s is more economical because it does not build causality out of a fermionic projector, it *starts* with the causal order.

8. The one open theorem (stated honestly, and sharpened to a measurement)

The continuum limit, stated as the sharp thing it is. The naive statement — “does the BST causal set converge to (3,1) Minkowski?” — hides the actual difficulty. The real obstruction is the Kleitman-Rothschild (KR) theorem: as the number of elements grows, *almost all* posets collapse to a rigid three-layer “pancake” (height 3, ordering fraction $r \sim 0.63$) that is not manifold-like at any dimension. Generic causal sets are KR pancakes, and manifold-like causets form a set of measure zero. So the honest hard theorem is not “reach (3,1)” — it is “escape the KR pancake”: show that BST’s forced growth law does *not* land in the generic non-manifold trap.

One half of this is now measured; the other half is not — and the two must not travel in the same sentence. The direct, runnable diagnostic at the physical size is the Myrheim-Meyer ordering fraction: KR pancakes sit at $r \sim 0.626$, while at $N=137$ the commit-forced order reads $r \sim 0.098$ — a clean $\sim 6x$ separation. Banked result: by the ordering fraction, the poset the commit operator $\exp(-\tau H_B)$ forces is NOT the Kleitman-Rothschild pancake — it escapes the generic measure-zero trap. (A caveat we owe, Elie 5261: the height* discriminator, natural as it seems, does not separate at the physical size — the commit order’s height is 3.8 vs KR’s 3 at $N=137$, too close. Height-growth is instead the continuum-limit leg — height grows unboundedly as $N \rightarrow \infty$ whereas KR stays flat-3 — an asymptotic statement, not the physical- N discriminator. The ordering fraction is the banked escape fact.)* This is real progress toward the continuum limit, and it is all that is measured.

What is NOT measured: the dimension. Escaping the KR pancake says the order is *not* the generic non-manifold object; it does not say the continuum dimension is 4. The Myrheim-Meyer ordering fraction (calibrated on Minkowski slabs, $r \sim 0.487 / 0.232 / 0.102 / 0.033$ at $d = 2-5$) is a dimension *estimator* only under a manifold-like, region-matched sampling that we have not yet established for the commit-forced poset at the required N . The dimension therefore remains open and unmeasured — the next step is a larger- N run with the region-matched interval estimator, not a larger claim. “Escapes KR” and “dimension is d ” are two different statements, and only the first is in hand.

What BST brings that generic causal-set theory does not: - the order is forced (not random) — the reason there is any hope of escaping a measure-zero trap: BST is one specific geometry-fixed poset, not a sample from the generic ensemble; - the tangent-space signature is (3,1) (Structure-Derived, T2545) — but this is the signature of the *tangent* algebra, not the continuum-limit dimension, which remains open and unmeasured; - the volume measure is given (commitment count) — Order + Number are both in hand.

So the open theorem has two parts, and only the first is closed: (i) *the commit-forced order escapes the KR pancake* — shown (ordering fraction 0.098 vs 0.626 at $N=137$); (ii) *the continuum limit is (3,1) Minkowski, at dimension 4* — open and unmeasured. BST enters the hard problem with the tangent signature and the volume measure pinned and the KR trap escaped, and the continuum dimension and metric still to be established.

8.1 The $5 \rightarrow 4$ descent is a FRAMEWORK-tier posit (exactly as GR posits its manifold)

The commitment order lives on the full 5-dimensional Shilov boundary of $D_{IV}^5 (S^4 \times S^1 / Z_2)$. The corpus carries a group descent to a 4-dimensional conformal boundary,

$SO(5,2) \supset SO(4,2) \supset SO(3,1)$ (Casey #14; T2545; F66): $D_{IV}^5 \rightarrow D_{IV}^4$, whose Shilov boundary is $R \times S^3 =$ conformally compactified Minkowski₄; a further descent to $SO(3,1)$ gives the tangent Lorentz signature (3,1).

This descent sits at FRAMEWORK tier — posited, not derived — exactly as general relativity posits its spacetime manifold. We investigated whether BST’s dynamics *forces* it, thoroughly (five numerical probes, one theorem, and three successive reframes each self-corrected). The honest result: it does not, and the descent lands at its floor unchanged. We state the ledger plainly, without gloss.

What the investigation established (real, but it did not move 3+1 closer): - No-go theorem (Elie 5257). BST cannot derive a preferred V_5 direction from single-valued $SO(5)$ -equivariant geometry — such a direction would be fixed by all of $SO(5)$, and the only $SO(5)$ -fixed vector in the vector rep is 0 (Schur). *Two scope caveats we keep sharp*: this is “cannot derive,” not “predicts no preferred frame exists” (GR is covariant yet has a CMB frame); and it concerns

the $SO(5)$ -internal direction V_5 , not $SO(3,1)$ spacetime covariance — the bridge between those is the descent itself. - Unique symmetric ground state (Elie 5259). The derived operator's ground state is unique and $SO(5)$ -invariant ($N = 2, 3, 4$) — no spontaneous ground-state degeneracy is exhibited. - Spectral matter is direction-blind (Elie 5258). The derived Dirac sea is $SO(5)$ -invariant. - Tangent $(3,1)$ is Structure-Derived (T2545) — the local-algebra signature, distinct from the global descent.

The missing piece is the bridge $SO(5)$ -internal $\rightarrow$ $SO(3,1)$ -spacetime, which is Casey #14 — and it sits at FRAMEWORK tier, posited, not derived. The no-go proves that single-valued equivariant geometry cannot promote it.

The last hatch — spontaneous symmetry breaking — is closed by finiteness. SSB is the one route the no-go does not cover, but it needs a genuine infinite-volume limit (so broken vacua superselect instead of tunnelling). BST does not have one *for the actualized content*: the last hatch is spontaneous symmetry breaking — and the honest question is not “finite or infinite” but *which sector*. BST is *both*: a finite interior (the committed record) and an infinite exterior (the ambient $H^2(D_{IV^5})$). Each is addressed: - Interior (finite) — no SSB. The committed record is finite: D_{IV^5} is bounded in $\mathbb{C}^5$, giving a finite spectral cap $N_{\max} = N_c^3 \cdot n_C + \text{rank} = 137$; the catalog is the finite field $GF(2^g) = GF(128)$; the substrate sums over a finite spectrum (“records fill toward the cap”, F865). A finite system has no infinite-volume limit — would-be-broken vacua tunnel and the ground state is unique and symmetric. The derived operator confirms it: the interior gap is 4.0000, flat across $N = 2, 3, 4$ — the $SO(5)$ vector Casimir, a *symmetry-protected* gap that does not soften toward degeneracy (Elie 5260). (*Scope, per Elie: N is a mode cut-off, not spatial volume — this closes the stated H^2 -infinite counter, not SSB in full generality.*) - Exterior (infinite) — a frame Machian-ly, no SSB needed. The infinite exterior sector *does* carry a frame — not from internal breaking, but because its actual contents do: the frame is the exterior's realized state, recorded on the boundary via light-absorption — a covariant boundary condition (Casey, K1522), the CMB-frame analogue. This subsumes the infinite sector: it has a rest frame because its matter does, exactly as in GR, and needs no SSB.

Status (Cal's verdict, rendered — #121): COMPLETE-AT-FLOOR. The $5 \rightarrow 4$ descent is a FRAMEWORK-tier structural posit, exactly as GR posits its spacetime manifold. Three closures stand — the geometric no-go (5257), the interior finiteness / symmetry-protected gap (5260), and the Machian exterior frame (K1522). Internal SSB is *unsupported*, not *excluded*: two premises (a dynamics and a derived superselection rule) remain un-exhibited, so the referee does not certify exclusion — only that the posit is complete at its floor with no supported route down. One named reopening condition: exhibit a genuine infinite-volume limit in which the symmetry-protected gap *closes* (the SSB entry). The theorem's lasting value: it permanently bounds the search space — a boundary marker that never has to be re-derived. What stands, banked and independent: the not-KR positive (Section 8; object confirmed), the causal-set structure, the derived *order*, Order + Number = Geometry, and the tangent $(3,1)$ signature.

Casey's override — the descent is NOT closed for us. The “complete-at-floor” verdict is the *referee's* status of the record; it is not an instruction to stop. Because BST is better-positioned than generic causal-set theory (the order is *forced*, not randomly grown; it escapes KR; the tangent signature is $(3,1)$), the manifold and the continuum limit are to be attempted, not parked — from D_{IV^5} and the corpus, via the *dynamical* route the no-go cannot touch (the continuum limit of the growing causal set $\rightarrow 4$ -D, whereupon the manifold emerges and the descent is realized dynamically). We package only at a genuine block, not from caution. This is the open frontier of Section 8, and it is where the work continues.

9. Status

- DERIVED / Structure-Derived: BST's causal structure is a locally finite partial order (causal set); the order is forced by the D_{IV^5} commit operator; the tangent-space signature is $(3,1)$ (T2545); the volume measure is the commitment count; the order is Lorentz-invariant with boosts as re-foliations (K1227).
- ESTABLISHED (imported): Malament-HKM (order $\Rightarrow$ conformal metric) and Sorkin (number $\Rightarrow$ volume) are standard results BST *feeds*, not claims.
- SHOWN (banked): the commit-forced order escapes the Kleitman-Rothschild pancake — at the physical size $N=137$ its Myrheim-Meyer ordering fraction is 0.098 vs KR's 0.626 (~6x, the discriminator; Elie 5261). (*The earlier “height grows” leg does not separate at $N=137$ — height 3.8 vs 3, too close; height-growth is the*

continuum-limit leg, $N \rightarrow \infty$, not the physical-size discriminator.) BST is not in the generic non-manifold trap.

- OPEN (unmeasured): the continuum-limit dimension and metric. Escaping KR is not a dimension measurement; the Myrheim-Meyer ordering fraction becomes a dimension estimator only under a region-matched manifold-like sampling not yet established at the required N . Next step: larger- N run with the region-matched interval estimator — not a larger claim.
- Not claimed: that the continuum limit is proved; that the continuum-limit dimension is 4 or (3,1) (only the *tangent* signature is pinned; the continuum dimension is unmeasured); that BST refutes CFS (it does not — it is the simpler substructure).

Registry: T2564. Companion (courtesy to the CFS community): the one-page Finster matrix (Cal) — “BST realizes causality via the commitment order, not Finster’s spectral A_{xy} : a different, simpler, self-standing route.”